

Simulation-based FABO first-aid Education: A Scoping Review

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Abstract

While simulation-based FABO (foreign body airway obstruction) emergency training has been common practice for medical professional, few studies have investigated the needs of the public first responders. To understand design trends, gaps and opportunities for simulation-based first aid training in this context, researchers conducted a scoping review across four databases and this process yielded 19 eligible papers that shared the design and application of simulation-based education in the FABO setting. Through this analysis, we understand the state-of-art in simulation content themes, design considerations and technologies discussed in these papers, identifying key research opportunities and challenges for the future. Dominant design strategies include (1) skill acquisition and decision making, (2) interactive procedure guidance, and (3) evaluation of the learning outcome. Key underlying design considerations are immersion and presence, as well as preparation emotion for the real-world scenes. Notably, knowledge retention, cognitive load, holistic scenarios, multi-sensory interaction are ought to be stressed for future layperson centred FABO simulation.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

virtual reality, mixed reality, simulation, pediatric airways, airway emergencies, medical education, choking, airway obstruction, first-aid

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1 Introduction

FABO (Foreign Body Airway Obstruction) are significant causes of injury and death among all age group especially in infants and young children with about 1,500 hospitalisations due to choking and suffocation in 2021-22, and young children (ages 0-4) having

the highest rates of these incidents [1]. Several reputable organisations provide practical guidelines and resources aimed at educating parents on this critical issue. Royal Australian College of General Practitioners (RACGP) developed detailed protocols and advice for preventing and managing airway obstructions in infants [22]. Simulation-based learning is a progressive educational approach that offers healthcare students and professionals expanded opportunities to develop clinical knowledge, skills, and attitudes [48]. By exposing trainees to a diverse range of procedures, this method allows for more consistent and standardised training within a speciality, facilitating faster learning without relying on the availability of patients [16]. The literature review aims to explore current practices in simulation-based training and identify design opportunities to extend FABO emergency simulation already widely used by medical professionals to laypersons including parents, caregivers, and first responders.

1.1 Research Questions

Despite thorough research, there appears to be a lack of comprehensive review studies specifically addressing state-of-the-art methods and the effectiveness of simulation based medical education on FABO. This scoping review's main goal is to identify the applications and design methodologies of simulation-based learning by examining research and design trends throughout the HCI literature. This emphasis on design is distinct from earlier literature studies that focused on the outcomes and feasibility of specific tools, such as virtual reality (VR) for FABO training in order to synthesise the current state of study. It is also worth noting that the majority of papers in the area focus on examining or designing general categories, such as CPR or BLS (Basic Life Support) (e.g., [17]), within which the state-of-arts may vary across different subjects. Besides, the research should consider how to support thoughtful and meaningful design innovation. Our scoping review explores the following research questions:

- Q1: What aspects were involved in the simulation part of content design?
- Q2: What are the underlying design considerations behind the simulation content?
- Q3: What are the factors that are assisted through simulation-based technology?

2 Related Works

Choking, asphyxiation, airway obstruction, and foreign body airway obstruction (FABO) [40] are often confused but represent distinct challenges in medical emergencies. **Choking** typically involves a foreign object blocking the airway, requiring immediate

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intervention to prevent asphyxiation. However, **airway obstruction** encompasses a broader range of blockages caused by factors such as swelling, trauma, or medical conditions [39]. **Foreign body airway obstruction (FABO)** was airway obstruction caused by a foreign object that becomes lodged in the respiratory tract, commonly in the upper airway and more often seen among paediatric population [45]. **Asphyxiation** (suffocation, asphyxia or suffocation) in medical context refers to a lack of oxygen that has a variety of causes including physical object blockage, external force, or drowning, etc [15]. Therefore in this paper we will use the term FABO to refer to the situation that caused by the obstruction of the airway due to a foreign object that partially or completely blocks airflow, often necessitating prompt unspecialised intervention to restore breathing.

Simulation can be categorised into human and non-human types. Human simulations involve role-playing among students, while non-human simulations utilise tools such as manikins or computer-based systems. Additional, simulations can be classified by type (compiler-driven or event-driven) or by fidelity level (low, medium, high) [19]. Simulation-based learning enhances situational awareness and decision-making in high-pressure environments, benefiting caregivers and first responders [43]. Its applications have shown educational benefits in recognising and managing incidents such as fires, car accidents, and construction site emergencies [50]. Haptic simulators are increasingly used in medical and emergency training, including firefighting scenarios [12]. BLS techniques like CPR are commonly incorporated into emergency education programs [41]. However, much of the literature emphasises technology over design. AR systems have demonstrated potential for visualising airway anatomy [27], improving users' kinaesthetic perception and fine motor skills. Immersive VR with haptics replicates real-life conditions, providing tactile feedback essential for medical procedures [23]. Haptic feedback, such as wristwatch force sensors, has been used to classify abdominal thrusts in choking first aid [52], while AR-based haptic simulations have been applied to infant airway obstruction training [31]. Despite these advancements, gaps remain in designing systems for caregivers and untrained bystanders. A 2024 review emphasises the importance of user-friendly haptic technologies to enhance accessibility and effectiveness [26]. Psychological factors such as panic and anxiety can hinder layperson performance in first aid procedures, such as the Heimlich manoeuvre or CPR [35]. A human-centred approach to healthcare simulation, grounded in human factors theory, has been advocated to prioritise user and community needs over purely technical advancements [42].

3 Methods

The JBI framework is used for the review to explore diverse concepts and settings within a broad topic across different types of studies and evidence without focusing on assessing quality or effectiveness [28]. A literature search was conducted in October 2024 using 4 databases; Google Scholar, Institute of Electronic Engineers (IEEE Xplore), PubMed, and Springer Link. These databases were selected based on prior experience and previous scoping reviews to capture a diverse range of publications in HCI and related fields. We used a combination of search terms and variations including

“virtual reality,” “VR,” “AR” “choking” “FABO” “Airway obstruction”, etc. The selected search terms were aligned with the research questions, aiming to retrieve articles that not only focused on VR experiences but also broad range of simulation forms. The process of study inclusion was carried out as follows: peer reviewed articles published in English after 2017 that described simulation-based FABO trainings (both customised or commercially available products) aimed at managing the physical, psychosocial design aspects for all age groups. The year 2017 marked a heightened use of head-mounted displays like the Oculus Rift in medical training [29]. Furthermore, publications were included regardless of locations of the trainings implemented, to include various settings such as hospitals, schools, or other types of care centres.

3.1 Study Selection and Evaluation

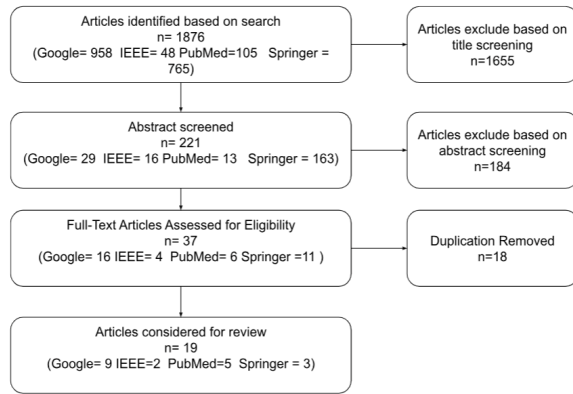
A total of 1876 articles were identified. The first author screened the titles and abstracts based on relevance and eligibility criteria, resulting in 19 articles included in the final shortlist for the scoping review. The elimination process is shown in Figure 1. All screened articles were analysed using a coding structure driven by the research questions, comprising three parts. The first part focuses on the simulation tasks and content explicitly included in the design. The second addresses design considerations - key elements impacting the educational experience, such as enhancing user presence through audio-visual mapping. The third explores influencing factors, such as technical or design issues, that affect learning outcomes and training effectiveness. The third part is simulation-based learning related technology. By reviewing the work of Brunzini et al, who proposed a design framework for transversal (e.g., emergency room) and specialised skills that included: **feature definition, tasks definition, feedback definition** in designing the element of simulating scenario. Each of the three parts would be designed if it be real, virtual, or both [10]. We aim to answer the research question (RQ3) by examining these three aspects. Apart from coding, a data collection sheet was also developed to collect details such as author, technology used, goals of study, design content, etc. Answering these research questions will allow us to identify further opportunities, which are outlined in the discussion section.

4 Findings

The term "FABO" was explicitly used in a few studies (n=2, 10.5%) [24, 38], while more papers used "choking" (n=7, 36.8%) [4, 9, 25, 30, 46, 49, 52], "airway obstruction" (n=5, 26.3%) [7, 14, 31, 32, 36, 37], "airway emergency" (n=1, 5.3%) [44], "new-born live training" (n=1, 5.3%) [47], and "obstructed respiratory tract" (n=1, 5.3%) [18]. Explicit references to FABO or related terms were key inclusion criteria, indicating the cause, treatment methods, and patient groups. Most studies reported positive learning outcomes, and comparative studies showed simulation-based learning outperforming other formats [21, 47, 49]. The most popular approach for FABO education combined AR/VR with physical manikins (n=6, 31.6%), followed by single-device use: VR (n=3, 15.8%), interactive or non-interactive manikins (n=3, 15.8%), and serious games (n=3, 15.8%). Studies using combined AR/VR and manikins often employed hand mapping technologies like Leap Motion or hand controllers with head displays. Other technologies included interactive manikins (n=3,

Table 1: Search terms used across five databases for the scoping review

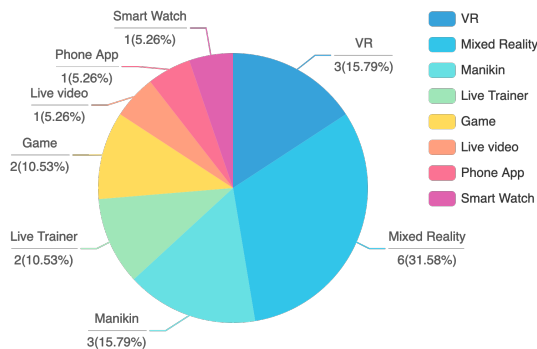
Database	Search Terms
Google Scholar	1. "choking" OR "airway obstruction" OR "airway emergencies" AND "first-aid" OR "treatment" OR "education" AND "virtual reality" 2. "choking" OR "suffocation" AND "first-aid" OR "treatment" OR "education" AND "virtual reality" OR "simulation" AND "haptic" OR "touch" OR "feedback" AND "back blow" OR "abdominal thrusts"
IEEE	(airway OR first-aid OR choking) AND (virtual reality OR simulation)
PubMed	(choking OR "airway obstruction") AND ("first-aid" OR education OR training) AND ("virtual reality")
SpringerLink	(choking OR "airway obstruction") AND ("first aid" OR education OR training) AND ("virtual reality" OR "mixed reality" OR "augmented reality" OR simulation)



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Figure 1: Flowchart of the literature screening and selection process

15.8%) [21, 47, 49], manikins with pressure sensors (n=1, 5.3%) [31], smartwatches with accelerometers (n=1, 5.3%) [52], smartphone applications (n=1, 5.3%) [32], games (n=2, 10.5%) [4, 9], and live video (n=1, 5.3%) [25]. A few studies relied on low-tech real-person acting with verbal instructions (n=2, 10.5%) [7, 36]. Details addressing the research questions are discussed in the following sections.

**Figure 2: Simulation technology used in the reviewed articles.**

4.1 Research Question 1: What aspects of simulation content design and underlying design considerations, contribute to effective first-aid training for FABO?

4.1.1 Scenario-based contextualisation. A common goal to design educational and training content is to enhance Immersion and Presence that has been mentioned by most of the papers as the preliminary designing consideration [7, 14, 47]. However this is different in the impact variables by choosing a VR or non-VR experience. VR simulations often adapt the strategy of visual contextualisation by replicate the scenarios such as operating room, delivery room [47]. The reason was typical expressed by enhancing engagement. AR was also used with pressure sensors to allow replicate realistic infant reactions [14]. Contextualisation is also used by giving tactile sensations in those using physical manikins and wearing apparatus[36]. In the study by Lv et al. [36], the participants expressed the preference for the live-actor wearing a apparatus with pressure sensors to evaluate performance than the VR training. Lv et al. [36] found VR had higher workload demands and less efficient gaze and motor patterns, leading to prolonged task completion times. Real-world tasks benefited from more systematic visual scanning and movement efficiency although both VR and real-world conditions elicited similar task behaviours (e.g., airway obstruction checks and compressions).

Many studies highlight emotion as a key design factor, often using varying terminology. Confidence-building is a common design goal, as seen in simulations for family members, which progress from basic tasks (e.g., handwashing, diaper changes) to complex skills like CPR. Structured scenarios, such as managing a stable baby or handling a fever, were designed to simulate familiar, real-life contexts [49]. Situational motivation is explicitly addressed in some studies [18], while others incorporate factors like noise and time pressure to simulate combat-like environments [36]. Most, however, use low-pressure settings focused solely on task performance [9, 31, 36]. Although not explicitly focused on stress, one study found that simulated death scenarios enhanced teamwork attributes such as communication, problem-solving, and professional relationships [30]. Another study emphasised the role of empathy in fostering better decision-making among first-aid responders [14]. Increasing bystander willingness to assist has also been explored, linked to situational motivation [18] and confidence in performance [32].

4.1.2 Skills Acquisition, Decision Making, and Procedure Guidance. Six of the articles evaluated the first-aid skill improvement of the participants after learning through the proposed simulation using pre-and-post test to measure knowledge improvement [9, 14, 32,

36, 49, 52]. Majority of the articles taught users to identify various symptoms through audio cues (gasping, choking noises), and body movements as the testing signs of airway obstruction that require actions accordingly in the give time. Decision making was also explicitly mentioned by some of the reviewed study [9, 14, 44]. The motivation behind could be reducing the time of hesitation in order to start the action as early. Comparing to other clues, facial expressions are relatively less considered part with only 5 among all article have included [4, 9, 18, 31, 46]. Also within the procedure, suffocation prevention is a less recognised area in many of the studies. Only 3 have designed recognition hazardous objects [9, 36, 46].

A considerable number of the studies (6 of 19 the papers) utilised different clues to guide users through the designed procedure. Procedure guidance were used as text and voice clues [31, 46, 47], real-time voice instruction [25], real-time visual overlay [31], app-based visual prompts [32], in-game visual and audio guidance [9] to guide users through sequences of actions and decisions. However, some relied on instructor observation and personal feedback [24, 38]. One introduced vibration feedback together with visual colour-coded signals for correct and wrong hand movement [52]. Some studies didn't include any in-action guidance in theoretical training [9, 44] but focused only on diagnostic simulation [37].

4.1.3 Evaluation of Learning Outcomes. Two of the studies stated the evaluation as the main design objective to dominantly explore the assessment to evaluate the learning outcome [4, 31]. the study from Alonso-Fernández et al. used designed a prediction tool for through game-based learning. Pao-Ju Chen and Wei-Kai Liou [31] tried to use AR system to record and evaluates the performance of learners in real time, when a user delivered back slaps or chest thrusts, the sensors sent electronic signals to the system. If the actions were performed correctly in terms of position, frequency, and strength, positive feedback appeared on the screen with a light activated. The AR environment allows learners to repeatedly practice first aid skills in a simulated, low-pressure setting. Apart from these, some studies embedded testing tools in the simulation experience [14, 32, 37, 44, 52] for the learning outcome during the evaluation process or incorporate self-evaluation in the simulation content [9, 14, 31, 52]. [46] generated score-based accuracy, speed, speed to quantify and task completion outcome.

4.1.4 Accessible and Adaptive Learning Environment.

Distant and independent learning is another consideration found. For example, It's used to create a self-directed and stress-free environment to enhance efficiency of distance- and self-learning by Pao-Ju Chen and Wei-Kai Liou [31]. One study by Arthur et al. [7] aimed to explore transfer effect of skills learnt through VR environment and the physical laboratory setting. Economy and adaptability was a consideration for the serious game [9].

4.2 Research Question 2: What are the factors that are assisted via simulation-based technology?

Interactive features encompass environmental characteristics (e.g., location, layout), clinical actors (e.g., profiles, roles), and patient anamnesis (e.g., scenarios for diagnosis and management). Most

studies utilised real or mixed environments for feature settings. Interactive tasks involve learner-patient interactions, considering verbal and physical communication, potential complications, clinical risks, and patient management. Feedback is provided through physical systems (e.g., manikin sounds, ECG signals) or virtual systems (e.g., holograms of patient parameters). While various sensors have been used to quantify actions and evaluate performance, multi-sensory feedback remains under-explored. Studies using MR technologies often employ haptic feedback from virtual systems rather than physical systems. In-action multi-sensory feedback offers subtle, real-time guidance, surpassing pre-practice verbal or video instructions [34]. However, some studies lack detailed descriptions of tasks or feedback [7, 32].

5 Opportunities and Challenges

5.1 General Gaps in simulation design for airway emergency research

Among the reviewed studies, it was found that traditional learning strategy was in dominance. There were no papers explicitly specify the learning theories used to guide design and application, advanced technology like VR or MR was used as a guidance system or as feedback tool.

The reviewed articles highlight that technology infrastructure often lacks ergonomic fidelity. Input devices with low precision and delays can hinder free hand movement in virtual spaces, impacting immersion and engagement [21, 44]. VR systems also face challenges like instability in virtual hand tracking, reducing interaction precision [21]. Additionally, VR sickness is noted as a potential barrier to learning [18]. While mixed reality often provides better tactile feedback and realism in scenarios like airway management, issues such as manikins failing to replicate loss of consciousness or lacking physical weight and detailed symptoms limit realism [24, 53]. Innovations to improve simulation usability in FABO education could include lighter, more comfortable head-mounted displays (HMDs), improved user interfaces, and binaural ambisonic audio [33, 48] to mitigate motion sickness. A VR prototype allowing users to practice recovery positions on a virtual mannequin [53] offers an example of managing ergonomic fidelity by bridging cognitive representation and motor skill execution.

HCI can enhance usability by addressing the needs of diverse user groups, including healthcare professionals, educators, and family caregivers. However, there is a notable gap in designing for parents and caregivers. Most studies focus on medical professionals and college students [7, 14, 18, 21, 25, 30–32, 36, 37, 44] or high school students [4, 9], with few targeting family parents [49], laypersons [24, 52], or children aged 10–13 [38]. Small sample sizes in many studies [7, 37, 44, 52] further limit generalisability. Compared to medical professionals, parents are more hesitant in emergencies involving their children, influenced by pressure and fear. Cognitive load assessment and prediction could better inform simulation designs to address these differences.

5.2 Design Opportunities

5.2.1 Multi-model Feedback System. It is a noticeable gap that feedback relating to the physical environment is under investigated among the reviewed studies. Future designs have the potential to

Table 2: Themes and design considerations identified

Articles	Themes	Design Considerations and Influencing Factors
Ricci et al. [47]	Identifying and responding to respiratory distress in newborns, immediate feedback on the accuracy and depth of chest compressions	Inclusive neonatal tool designed for all medical staff, immersion, knowledge
Girau et al. [21]	Not specified: general emergency first-aid tasks to tasks.	Reproduce real conditions, create contextualisations immersion and presence
Putnam et al. [44]	Evaluation of Learning Outcome	Decision making
Boada et al. [9]	Skills acquisition and Decision Making	Promote the first-aid procedure, low-cost game model
Pao-Ju Chen and Wei-Kai Liou [14]	Skills acquisition and Decision Making, Evaluation of Learning Outcome	record, quantify, distance training, cultivate empathy
Lauren A. Mansour [37]	A slide-based presentation that discussed diagnosing patients with airway or breathing complication	Enhance the detection and identification of subtle visual cues, AR vs. textbook
Brenda M. Seegmiller [49]	Skills acquisition and Decision Making	Confidence building
Amanda Watson and Gang Zhou [52]	Not specified, in-situ instruction and real-time feedback	Accurately classify back blows, combat the bystander effect, skill retention, video vs. watch application
Rainer et al. [46]	General first-aid procedures completion and earn scores	Realism and interactivity
Khattak et al. [30]	Skills acquisition and Decision-Making: Identifying and prioritising interventions for a critically ill patient. Deciding on the sequence and type of procedures based on the scenario.	Teamwork ethics, emotional maturity, and the ability to function under critical conditions. facing imminent death vs. regular living patient
Alonso-Fernández et al. [4]	Skills learning through self testing	Knowledge change based on in-game actions
Pao-Ju Chen and Wei-Kai Liou [31]	Evaluation of Learning Outcome	Apparatus for layperson lack of medical background, VR vs. living actor with apparatus
Lv et al. [36]	Evaluation of Learning Outcome	Develop assessment for layperson
Hasselager et al. [24]	Skills acquisition and Decision Making, first-aid performance, identification of subtle medical cues in diagnosing airway obstruction.	
Kim et al. [32]	Evaluation of Learning Outcome, Recognition and management of infant airway obstruction. Hands-on training for back slaps, chest thrusts, and CPR for unresponsive infants.	Knowledge, skills, confidence, phone app vs. lecture
Igarashi et al. [25]	Evaluation of Learning Outcome	Increase bystander effort, immediate management
Martínez-Isasi et al. [38]	Procedural skills, task motivation, and perceptual abilities.	Transfer effect - simulation validity and fidelity, VR vs. real-world synthetic environment
Arthur et al. [7]		Situational motivation, VR sickness, presence, usability, VR vs. MR with haptic reference
Elsenbast et al. [18]	Fast decision making, pre-hospital emergency treatment	

Table 3: Overview of interactions categorised

Interaction	Real	Virtue	Both
Features	[30, 36, 38, 49, 52]	[9, 32, 38, 46]	[14, 18, 21, 31, 37, 44, 47]
Tasks	[7, 30, 31, 36]	[9, 37, 46]	[14, 18, 21, 52]
Feedback	[30, 36, 38, 49]	[9, 14, 21, 31, 37, 44, 46, 52]	[18]

explore more sensory models to create rich and initiative feedback with various actuators. However, greater interaction richness does not always lead to better learning outcomes, and technology adaptability remains a challenge. Lv et al. found that live-actor training significantly improved first-aid skills in non-medical students compared to high-fidelity VR simulators, particularly in tactical injury assessment [36]. Another study using gaze data to compare VR and real-world practice reported higher perceived workloads and longer task completion times in VR, as learners processed virtual sensory cues differently from real-world cues [7]. This suggests that high-tech simulations may lack physical and ergonomic fidelity. Similar to the feedback system, the combination of interactive virtue and real tasks is also less considered. In this regard, using gamification could offer an interesting and hands-on learning experience that improves memory and application of life-saving abilities, making them a realistic and successful method for teaching non-experts the fundamental first aid protocols associated with choking situations [51]. Despite game elements such as stories and narratives, quizzes has been used in the reviewed studies, future research could further explore practical applications like remote collaboration, progression tracking, leaderboards to balance between enjoyable and

educational content [51] and incite both intrinsic and extrinsic motivation for individuals over the short, medium, and long term [54].

5.2.2 Design evaluation - from performance testing to long-term experience. A key gap in most studies is the lack of focus on long-term knowledge retention, a critical component of emergency training. VR methodologies showed similar retention declines to traditional methods, with increased competency immediately after training but decreased retention over time [13]. A study on choking knowledge retention found poor performance on key skills, such as hand placement and abdominal thrusts, regardless of time since training, and skill deterioration in compressions after 30 days [5]. These findings suggest that frequent certification is insufficient for retention, highlighting the need for methods to reinforce and refresh skills. Additionally, first aid knowledge assessed through multiple-choice exams was higher among those with advanced training and remained stable over time. This indicates potential for designs that provide a deeper and broader knowledge base for sustained understanding.

5.2.3 Physiological design for Stress Management. Although the review did not include the technical research by Appel et al. [6], their multi-modal approach using eye tracking, heart rate, and in-game activity to classify cognitive load offers potential for adaptive systems in emergency training. Similarly, integrating stress management into training could be beneficial. A Stress Reflection System for paramedic trainees, incorporating synchronised physiological and behavioural data, demonstrated how team-based reflection enhances emotional intelligence, communication, and stress awareness [2]. While higher levels of transformative reflection were not evaluated [20], these frameworks could also benefit the general public. Future research should consider cognitive load management strategies, such as spatial integration, visual attributes, and content segmentation, which have been shown to reduce visual attention changes in extended reality environments [8]. Additionally, strategies to mitigate sensory discomfort in FABO training and the use of advanced technology as guidance or feedback tools merit further exploration [11].

5.2.4 Situational life-connecting learning experience. Situational, life-connected learning remains underutilised, as most studies emphasise controlled environments with fixed time frame and settings, often making training feel detached from everyday life. Remote and mobile learning tools offer potential for situational learning by providing customisable and detailed virtual environments for skill refinement and real-world application. Features such as adjustable levels of distraction or intensity can create to diverse training needs. A first-aid VR training system (e.g., heatstroke, shock, and seizure) has also highlighted gender differences in learning interests [3]. To expand professional training applications, researchers should explore person-centred designs that integrate familiar environments and promote situated learning through collaborative group activities among diverse participants, regardless of age or BLS knowledge, to better reflect the unpredictability of real-life scenarios.

6 Conclusion

The use of only four databases may have excluded studies from other fields or regions. Additionally, this review omitted studies that did not explicitly address FABO or airway obstruction, potentially overlooking research on simulation-based learning for broader emergency scenarios or medical interventions. Future research could expand the inclusion criteria to address these limitations.

FABO first-aid, an essential aspect of emergency medical training, plays a crucial role in preventing choking-related fatalities and improving patient safety in healthcare education. Studies show that VR-based FABO training effectively improves knowledge, confidence, and response accuracy in real-life emergencies. This scoping review explored the state-of-the-art in this field and investigate how technology can support first-aid education. Using four databases, 19 articles were analysed across three themes: first-aid learning content in simulations, design considerations (factors influencing content design supported by technology), and simulation-related technologies such as AR, VR, and haptics. The findings highlight significant advancements in simulation-based learning for emergency first-aid, particularly in integrating tangible interaction with virtual objects in immersive environments. Future work should

focus on aligning processes, user groups, and learning objectives with appropriate technologies to maximise their potential.

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